Competitive Programming

Modulo Arithmetic

Modulo Arithmetic

- Some real-world artifacts exhibit wrap-around behavior
  - Time
  - Angle
  - Games / Simulations
- Demonstrate a large computation without requiring a (large) exact answer

Modulo Arithmetic

- $(a + b) \% m = ((a \% m) + (b \% m)) \% m$

If you need the sum of two (large) numbers modulo some number m.

You can apply the mod operation before you add.

Modulo Arithmetic

- $(a + b) \% m = ((a \% m) + (b \% m)) \% m$
  - Example
    - $(11 + 29) \% 7 = 40 \% 7 = 5$
    - $((11 \% 7) + (29 \% 7)) \% 7 = (4 + 1) \% 7 = 5 \% 7 = 5$
  - Example
    - $(47324983290527 + 573892829104573) \% 7 = 621217812395099 \% 7 = 4$
    - $((47324983290527 \% 7) + (573892829104573 \% 7)) \% 7 = (5 + 6) \% 7 = 11 \% 7 = 4$

Modulo Arithmetic

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Don’t compute this.

Compute this instead.

Modulo Arithmetic

- (a + b) % m = ((a % m) + (b % m)) % m
- (a - b) % m = ((a % m) - (b % m)) % m
- (a * b) % m = ((a % m) * (b % m)) % m
- Example:
  - (437284927 * 3724849384 - 428492374 * 423837284) % 7 =
    1447209446957562752 % 7 =
    2
  - ((437284927 % 7) * (3724849384 % 7) % 7 – (428492374 % 7) * (423837284 % 7) % 7) % 7 =
    (2 * 4 % 7 - 2 * 3 % 7) % 7 =
    (8 % 7 – 6 % 7) % 7 =
    (1 – 6) % 7 =
    -5 % 7 =
    2 % 7 =
    2

Modulo Arithmetic

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    (8 % 7 – 6 % 7) % 7 =
    (1 – 6) % 7 =
    -5 % 7 =
    2 % 7 =
    2

This could be a problem.

Modulo and Negative Values

- Modulo can behave differently, depending on language

```
printf( "%d\n", -19 % 7 );
```

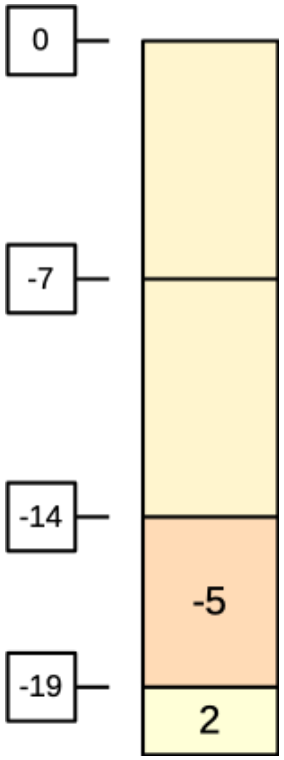
→ `-5`

```
System.out.println( -19 % 7 );
```

→ `-5`

```
print( -19 % 7 )
```

→ `2`



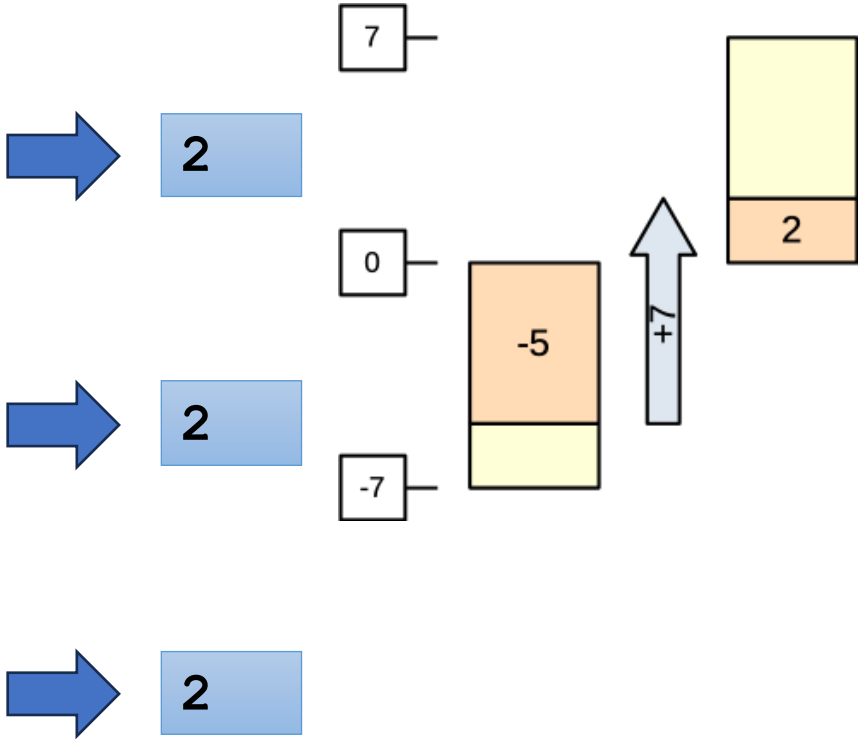
Modulo and Negative Values

- You can fix modulo, with a little work.

```
v = -19;
m = 7;
printf( "%d\n", ( v % m + m ) % m );
```

```
v = -19;
m = 7;
System.out.println( ( v % m + m ) % m );
```

```
v = -19
m = 7
print( -19 % 7 )
```



Modulo Arithmetic

- $(a + b) \% m = ((a \% m) + (b \% m)) \% m$
- $(a - b) \% m = ((a \% m) - (b \% m)) \% m$
- $(a * b) \% m = ((a \% m) * (b \% m)) \% m$
- $(a / b) \% m = ((a \% m) / (b \% m)) \% m$ … ?

Modulo Arithmetic

- (a + b) % m = ((a % m) + (b % m)) % m
- (a - b) % m = ((a % m) - (b % m)) % m
- (a * b) % m = ((a % m) * (b % m)) % m
- (a / b) % m = ((a % m) / (b % m)) % m … ?
- Example:
  - (34 / 16) % 7 =
    2 % 7 =
    2
  - ((34 % 7) / (16 % 7)) % 7 =
    (6 / 2) % 7 =
    3 % 7 =
    3

Oops. We didn't get the same result.

Modulo Arithmetic

- (a + b) % m = ((a % m) + (b % m)) % m
- (a - b) % m = ((a % m) - (b % m)) % m
- (a * b) % m = ((a % m) * (b % m)) % m
- (a / b) % m ≠ ((a % m) / (b % m)) % m

Modular Multiplicative Inverse

- We can't use regular division
- … but, if m is prime …
- … most values b have a multiplicative inverse module m.
- We'll write it as $\mathbf{b^{-1}}$
  - For any value a
  - $a * b * b^{-1} \% m = a$

Modular Multiplicative Inverse

- How do we find b^{-1}
  - We could search for it.
  - Consider b = 3 and m = 7

Modular Multiplicative Inverse

- How do we find b^{-1}
  - We could search for it.
  - Consider b = 3 and m = 7
  - 3 * 1 % 7 = 3

Modular Multiplicative Inverse

- How do we find b^{-1}
  - We could search for it.
  - Consider b = 3 and m = 7
  - ~~3 * 1 % 7 = 3~~
  - 3 * 2 % 7 = 6

Modular Multiplicative Inverse

- How do we find b^{-1}
  - We could search for it.
  - Consider b = 3 and m = 7
  - ~~3 * 1 % 7 = 3~~
  - ~~3 * 2 % 7 = 6~~
  - 3 * 3 % 7 = 2

Modular Multiplicative Inverse

- How do we find b^{-1}
  - We could search for it.
  - Consider b = 3 and m = 7
  - ~~3 * 1 % 7 = 3~~
  - ~~3 * 2 % 7 = 6~~
  - ~~3 * 3 % 7 = 2~~
  - 3 * 4 % 7 = 5

Modular Multiplicative Inverse

- How do we find b^{-1}
  - We could search for it.
  - Consider b = 3 and m = 7
  - ~~3 * 1 % 7 = 3~~
  - ~~3 * 2 % 7 = 6~~
  - ~~3 * 3 % 7 = 2~~
  - ~~3 * 4 % 7 = 5~~
  - 3 * 5 % 7 = 1

That's it right there.

Modular Multiplicative Inverse

- How about for larger values of m?
  - Like b = 32848925 and m = 33294841
- For prime m, we're guaranteed that $b^{-1} = b^{m-2} \% m$

Can we compute this?

Modular Multiplicative Inverse

- How about for larger values of m?
  - Like b = 32848925 and m = 33294841
- For prime m, we're guaranteed that $b^{-1} = b^{m-2} \% m$

b^{m-2} could be a really big value.

We could mod as we multiply rather than just at the end.

Modular Multiplicative Inverse

- How about for larger values of m?
  - Like b = 32848925 and m = 33294841
- For prime m, we're guaranteed that $b^{-1} = b^{m-2} \% m$

Can we afford m multiply operations?

We can do exponentiation by squaring … O(log m) time.

Modular Multiplicative Inverse

- How about for larger values of m?
  - Like b = 32848925 and m = 33294841
- For prime m, we're guaranteed that $b^{-1} = b^{m-2} \% m$
- Let's try this for b = 3 and m = 7
  - $b^{-1} = 3^{m-2} \% m = 3^5 \% 7 = (3 * 3 * 3 * 3 * 3) \% 7 =$
    $243 \% 7 = 5$

Modular Multiplicative Inverse

- How about for larger values of m?
  - Like b = 32848925 and m = 33294841
- For prime m, we're guaranteed that $b^{-1} = b^{m-2} \% m$
- Let's try this for b = 172 and m = 347
  - $b^{-1} = 172^{345} \% 347$
  - $172^{256} * 172^{128} * 172^{64} * 172^{32} * 172^{16} * 172^{8} * 172^{4} * 172^{2} * 172^{1} \% 347$

101011011
in binary